

Chemical
AMMONIUM IRON(II) SULFATE
hexahydrate

Prepared 01/12/01
By Dr. B. Dunlevy
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Section 1. Chemical identification

Name: Ammonium iron(II) sulfate hexahydrate

Section 2. Composition/information on ingredients

CAS #: 7783-85-9

MF: $\text{H}_8\text{FeN}_2\text{O}_8\text{S}_2$ or $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$

Synonyms

Ammonium ferrous sulfate hexahydrate; ferrous ammonium sulfate, Hexahydrate; iron ammonium sulphate hydrate; Mohr's salt; sulphuric acid, ammonium iron(2+) salt, hexahydrate;

Section 3. Hazards identification

Label precautionary statements

- Irritant
- Irritating to eyes, respiratory system and skin.
- In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
- Wear suitable gloves and eye/face protection.
- Air and light sensitive



Irritant

Section 4. First aid measures

In case of eye contact, immediately flush eyes or skin with copious amounts of water for at least 15 minutes. Assure adequate flushing of the eyes by separating the eyelids with fingers, and seek medical advice.

In case of skin contact, immediately flush skin with copious amounts of water for at least 15 minutes while removing contaminated clothing and shoes.

If inhaled, remove to fresh air. If not breathing give artificial respiration. If breathing is difficult, give oxygen.

If swallowed, wash out mouth with water provided person is conscious. Call a physician.

Wash contaminated clothing before reuse.

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Section 5. Fire fighting measures

Extinguishing media

Non-combustible.

Use extinguishing media appropriate to surrounding fire conditions.

Special fire fighting procedures

Wear self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes.

Unusual fire and explosions hazards

Emits toxic fumes under fire conditions.

Section 6. Accidental release measures

- Wear self-contained breathing apparatus, rubber boots and heavy rubber gloves.
- Sweep up, place in a bag and hold for waste disposal.
- Avoid raising dust.
- Ventilate area and wash spill site after material pickup is complete.

Section 7. Handling and storage

Store in **GREY** area.

Refer to Section 8.

Section 8. Exposure controls/personal protection

- Wear approved respirator EN149, chemical-resistant gloves, safety goggles, other protective clothing.
- Safety shower and eye bath.
- Mechanical exhaust required, use in the fume cupboard.
- Do not breathe dust.
- Avoid contact with eyes, skin and clothing.
- Avoid prolonged or repeated exposure.
- Wash thoroughly after handling.
- Irritant.
- Keep tightly closed.
- Air and light sensitive

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-
- Store in a cool dry place.



Goggles



Gloves



Respirator



Protective
clothing



Wash
hands

Section 9. Physical and chemical properties

Appearance and odour

Pale green crystalline powder

Physical properties

Specific gravity: 1.864

melting point: 100°C (dec)

Section 10. Stability and reactivity

Incompatibilities

Strong acids, air sensitive, light sensitive, strong oxidizing agents.

Hazardous combustion or decomposition products

Ammonia, sulphur dioxide and sulphur trioxide.

Section 11. Toxicological information

Acute effects

- May be harmful by inhalation, ingestion, or skin absorption.
- Causes eye and skin irritation.
- To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

Toxicity data

orl-rat LD₅₀:3250 mg/kg

Section 12. Ecological information

Data not yet available.

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Section 13. Disposal considerations

- Ammonium iron(II) sulfate should be dissolved in:
 - 1) water;
 - 2) acid solution or
 - 3) Oxidised to a water-soluble state.Precipitate the material as the sulphide, adjusting the pH of the solution to 7 to complete precipitation. Filter the insolubles and dispose of them in a hazardous- waste site. Destroy any excess sulphide with sodium hypochlorite. Neutralise the solution before flushing down the drain.
- Observe all EU, Irish and Local Environmental Regulations.

Section 14. Transport information

Contact chemical supplier for transportation information.

Section 15. Regulatory information

European information:

Caution: substance not yet fully tested.

Irritant

- R 36/37/38** Irritating to eyes, respiratory system and skin.
S 26 In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S 37/39 Wear suitable gloves and eye/face protection.



Irritant

OEL-IRL: TWA 1 mg (Fe)/m³; STEL 2 mg (Fe)/m³ 1999.

Section 16. Other information

This MSDS has been prepared in DkIT and is designed for Irish Second Level School Science Laboratories use only where quantities handled are less than 50g. The above information is believed to be correct but does not claim to be all inclusive and shall be used only as a guide. Every effort has been made to ensure that this information provided conforms with the latest available data. DkIT, the Department of Education & Science, Limerick Education Centre, Sigma, Aldrich, Fluka, Dr. Dunlevy or Ms. F. Dunlevy shall not be held liable for any information errors in this MSDS or for any damage resulting from handling or from contact with the above product.

Please address any comments or error observations to: brian.dunlevy@dkit.ie

REFERENCES

Sigma Aldrich Fluka MSDS CD-ROM, Sax CD-ROM, MSDS on Internet